$\{\ \}$

$$\frac{1}{2i}\left[-\eta^o+(\eta^o)^*\right]pu^* - \frac{1}{2i}\left[\eta^e+(\eta^e)^*\right]pu^*$$

$$+\ \frac{1}{2i}\left[(\eta^o)^*-\eta^o\right]p^*u + \frac{1}{2i}\left[\eta^e+(\eta^e)^*\right]p^*u.$$

$$-\operatorname{Im}(\eta^o)pu^* + \operatorname{Re}(\eta^e)\,i\,pu^*$$

$$-\operatorname{Im}(\eta^o)p^*u - \operatorname{Re}(\eta^e)\,i\,p^*u.$$

$$-\operatorname{Im}(\eta^o)(pu^*+p^*u) + \operatorname{Re}(\eta^e)\,i\,(pu^*-p^*u.)$$

$$pu^* + p^*u = 2\operatorname{Re} pu^*$$

and

$$pu^* - p^*u = 2i\operatorname{Im} pu^*.$$

So we have inside $\{\ \}$ s:

$$-2\operatorname{Im}(\eta^o)\operatorname{Re}(pu^*) + 2\operatorname{Re}(\eta^e)\,i\,i\operatorname{Im}(pu^*)$$